

Boating Trauma

Section 1: Case Summary

Scenario Title:	
Keywords:	Boating accident
Brief Description of Case:	25 year old male is involved in a boating accident. He was in the lake waterskiing and struck by another boat, sustaining blunt abdominal and pelvic injuries and significant propeller injuries to his leg. The trauma team will be activated and he will be brought to the ED. Treatment includes tourniquet application to the limb and rapid blood product administration. Intubation for clinical course and pain management need to be addressed with disposition to the operating room.

Goals and Objectives	
Educational Goal:	To enhance resuscitation and team management skills through a trauma scenario with multiple key interventions.
Objectives: (Medical and CRM)	1) To work as an interdisciplinary team to effectively manage a polytrauma patient 2) To review overall management of a polytrauma patient, including tourniquet application and blood product administration 3) To evaluate workflow process in the polytrauma patient with exsanguinating limb injuries and blunt abdominal trauma
EPAs Assessed:	F1: Initiating and assisting in resuscitation of critically ill patients C2: Resuscitating and coordinating care for critically injured trauma patients C5: Identifying and managing patients with emergent medical or surgical conditions C13: Performing advanced procedures

Learners, Setting and Personnel				
Target Learners:	☐ Junior Learners		☐ Senior Learners	
	☐ Staff			
	☐ Physicians	☐ Nurses	☐ RTs	☒ Inter-professional
	☐ Other Learners:			
Location:	☐ Sim Lab		☒ In Situ	
	☐ Other:			
Recommended Number of Facilitators:	Instructors: 1			
	Sim Actors: 0			
	Sim Techs: 1			

Scenario Development	
Date of Development:	30/05/2024
Scenario Developer(s):	JoAnne Slinn
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Last Revision Date:	
Revised By:	
Version Number:	



Boating Trauma

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Steve		Age: 25		Gender: M	Weight: 70kg
Presenting complaint: Boating trauma					
Temp: 36.0	HR: 120	BP: 90/60	RR: 24	O ₂ Sat: 96%	FiO ₂ : RA
Cap glucose: 6.2			GCS: (E V M) 14 (E4 V4 M6)		
Triage note: 25 year-old male was involved in a boating accident. He was waterskiing and was struck by another boat at high speeds. He sustained significant propeller injuries to right leg, unstable pelvis, and abdominal bruising. No LOC but altered with some reported alcohol consumption. Was wearing a life jacket with minimal submersion. He was quickly pulled out of the water by his friends with significant blood loss on scene. EMS applied combat application tourniquet to leg on scene 45 minutes ago. Pelvic binder applied, spine stabilized, and IV established in Rt AC with 1L NS infused. TXA 2 gm given and methoxyfluorane for pain control enroute.					
Allergies: None					
Past Medical History: None			Current Medications: None		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, sim actors, SP, etc.)?</i>	

B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal S1/S2. No murmurs.	Neuro: Awake, slightly confused, screaming in pain
Resp: AE clear and equal to bases	Head & Neck: Unremarkable
Abdo: Diffuse bruising, tender, distended. Pelvis unstable	MSK/skin: Right lower leg mangled, lacerations, significant bleeding
Other:	



Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Tourniquet – manual and automatic Pelvic binder Blood
C. Required Medications
RSI medications Pressors
D. Moulage
Right lower leg mangled with significant bleeding Abdominal bruising
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A – patent, screaming in pain until analgesia provided B - tachypnea, lungs clear C – hypotensive, tachycardic, exsanguinating Right lower limb that is pulseless when tourniquet applied, abd tender with bruising, pelvis unstable D – Awake, slightly confused, screaming in pain, PERL, moving all limbs except Right leg E – No other injuries found, wet clothes removed

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)

Boating Trauma

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus tach HR: 120 BP: 90/60 RR: 24 O ₂ SAT: 96% RA T: 36.0°C GCS: 14 (E4 V4 M6)	Slightly confused, screaming in pain	<u>Expected Learner Actions</u> ☐ EMS handover ☐ Monitors, vitals ☐ Primary survey ☐ Inspect tourniquet to ensure bleeding controlled ☐ Bind pelvis if not yet done ☐ Obtain 2 nd IV access - <i>if unable then IO into unaffected limb</i> ☐ Call for blood and give rapidly ☐ FAST (+free fluid) ☐ Provide analgesia ☐ Call for Xray (CXR, pelvis, leg) ☐ Administer TXA IV ☐ Check cap glucose (6.2mmol/L) ☐ Blood work trauma panel (inc. G&S)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - No blood by 5 minutes → BP 70/30, HR 140 - Screaming in pain until analgesia provided <u>Triggers</u> <i>For progression to next state</i> - Actions complete → 2. Remains Hypotensive	Ongoing bleeding from leg that is controlled with tourniquet
2. Remains Hypotensive Rhythm: Sinus tach HR: 130 BP: 80/55 RR: 24 O ₂ SAT: 95% RA GCS: 13 (E3 V4 M6) post analgesia		<u>Expected Learner Actions</u> ☐ Activates MHP ☐ Switches to automatic tourniquet ☐ Sets up for intubation for OR ☐ Consult surgery	<u>Modifiers</u> - Resuscitates before intubating: 4U PRBCs → BP 100/60, HR 110 - No resuscitation, RN to prompt, "should we give more blood before we intubate"? <u>Triggers</u> - 4U PRBCs, intubated, surgery called → 3. Resolution -	



Boating Trauma

3. Resolution Rhythm: Sinus tach HR: 110 BP: 90/65 RR: 12 on vent O ₂ SAT: 98% ETT GCS: 3 (post induction)		<u>Expected Learner Actions</u> ☐ Post intubation sedation/analgesia ☐ Antibiotics & Tetanus ☐ OR ready – prepare to transfer	<u>Modifiers</u> - <u>Triggers</u> - OR ready → End Case	
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Appendix A: Laboratory Results

VBG

pH 7.12

Bicarb 14 mmol/L

pCO₂ 28 mmHg

Lactate 6.9 mmol/L

Appendix B: ECGs, X-rays, Ultrasounds and Pictures - *options*

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!

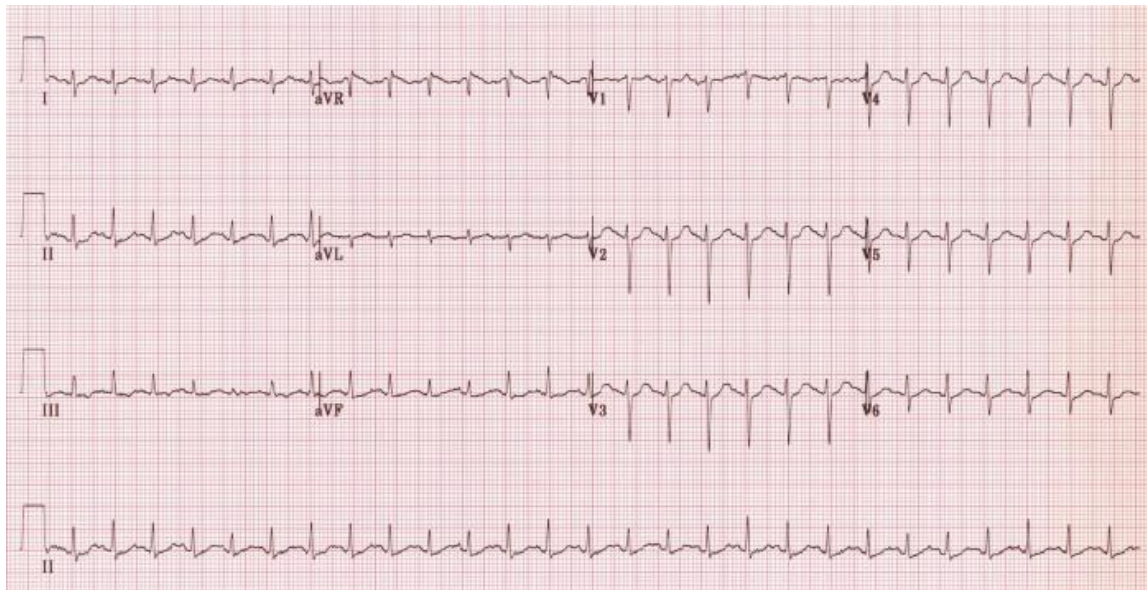


[Motorboat Propeller Injuries; A Case Series and Review of the Literature \(zgt.nl\)](#)

Boating Trauma



https://www.reddit.com/r/medizzy/comments/wbx2wb/boat_accident_causing_a_severe_leg_injury_the/?rdt=35419&force_seo=1

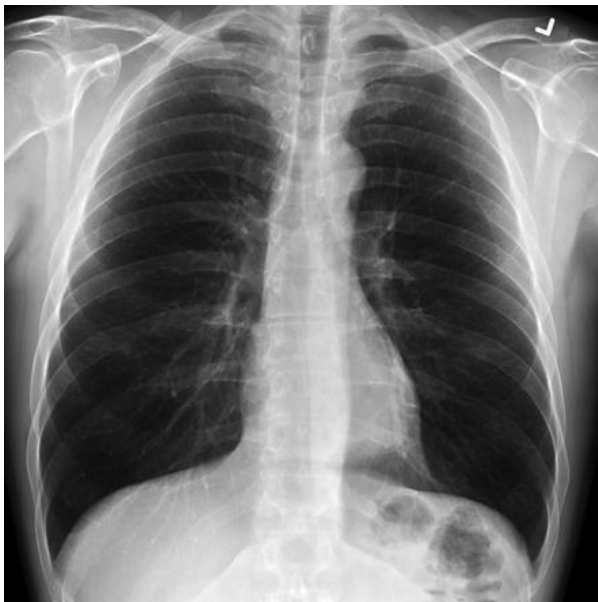


ECG source: <https://i0.wp.com/lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg>

Boating Trauma



[Trauma — TPA \(thepocusatlas.com\)](https://thepocusatlas.com)



(CXR source: <https://radiopaedia.org/cases/normal-chest-x-ray>)

Boating Trauma



(PXR source: <https://radiopaedia.org/cases/open-book-pelvic-fracture-4>)

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

STOP THE BLEED RESOURCE

APPLYING A TOURNIQUET

If you do have a trauma first aid kit:

For life-threatening bleeding from an arm or leg and a tourniquet is available:

■ Apply the tourniquet

1. Wrap the tourniquet around the bleeding arm or leg about 2 to 3 inches above the bleeding site (*be sure NOT to place the tourniquet onto a joint – go above the joint if necessary*)



2. Pull the free end of the tourniquet to make it as tight as possible and secure the free end

3. Twist or wind the windlass until bleeding stops



4. Secure the windlass to keep the tourniquet tight

5. Note the time the tourniquet was applied

Note: A tourniquet will cause pain but it is necessary to stop life-threatening bleeding.

Boating Trauma

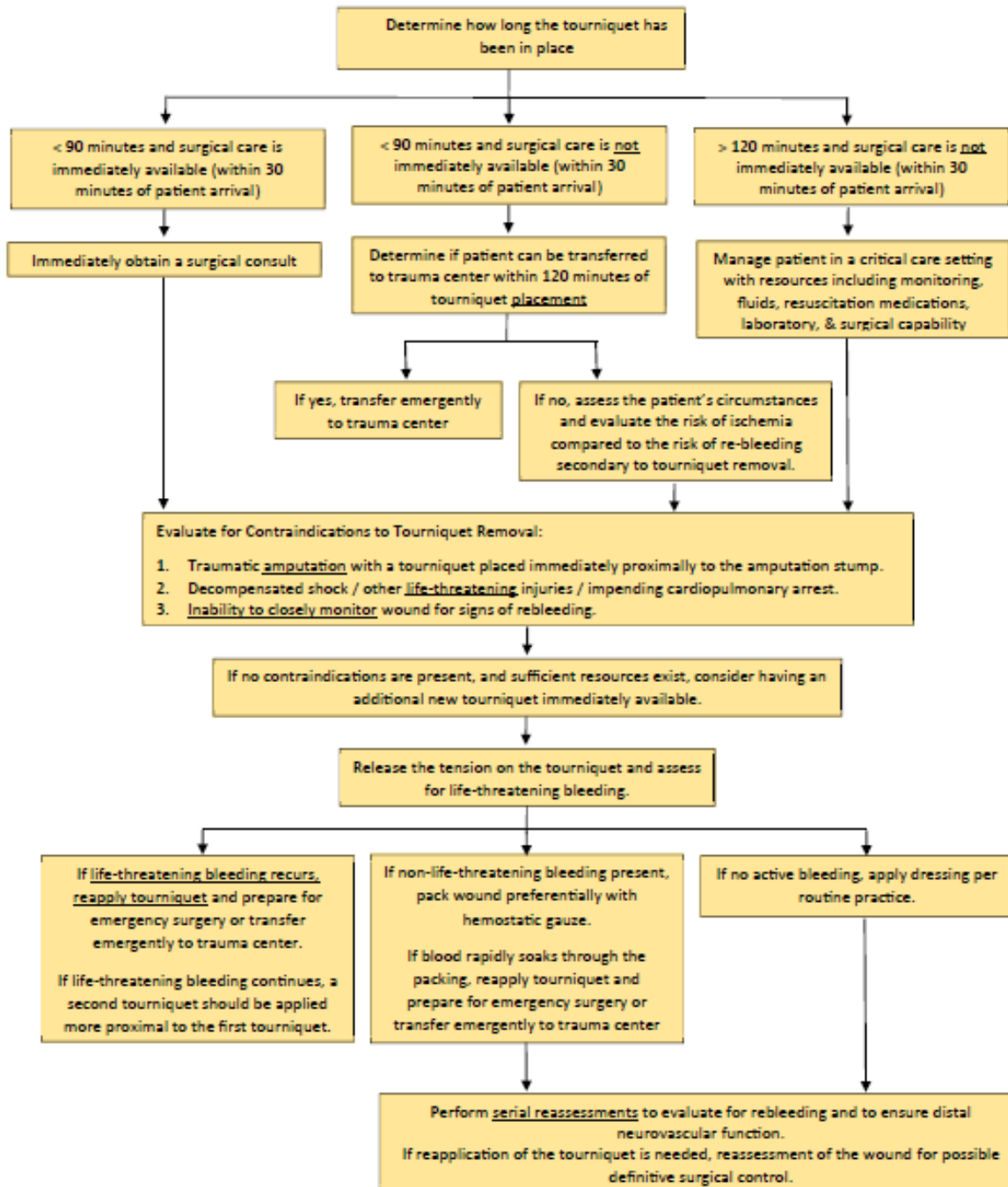


Figure 1. Emergency department approach to the patient with a prehospital tourniquet.

<https://pubmed.ncbi.nlm.nih.gov/33303278/>

References

Boating Trauma

1. <https://www.hilarispublisher.com/open-access/motorboat-propeller-injuries-report-of-thirteen-cases-with-review-of-mechanism-of-injury-and-bacterial-considerations-2167-1222-1000267.pdf>
2. [Motorboat Propeller Injuries; A Case Series and Review of the Literature \(zgt.nl\)](#)
3. <https://pubmed.ncbi.nlm.nih.gov/33303278/>

